ISAR TARGET PARAMETER ESTIMATION WITH APPLICATION FOR AUTOMATIC TARGET RECOGNITION

Kenneth A. Melendez and John R. Bennett
Science Applications International Corporation
10260 Campus Point Drive, MS C4, San Diego, CA 92121

ABSTRACT

Algorithms for estimating the angular velocity, angular acceleration, orientation, length, and width of an ISAR target and for identifying the scattering points of ISAR targets are derived and demonstrated. A correspondence between the ISAR motion estimation operations and a rigid body dynamical model is illustrated. It will be shown that an ISAR image amounts to projecting scatterers into a plane perpendicular to the target axis of rotation and coplanar with the range axis. The proper Doppler scale factor of imagery and its variation is proportional to the rotation rate and acceleration. These result as a by-product of SAIC ISAR motion estimation techniques. Improved geometric estimates can then be found by incorporating modeling assumptions and a priori knowledge. A number of land and sea examples are provided to reinforce the above concepts.

Keywords: *ISAR, Radar detection, Automatic Target Recognition*

1.0 INTRODUCTION

Inverse Synthetic Aperture Radar (ISAR) imaging differs from traditional SAR based upon the use of target motion to produce cross-range resolution. Most ISAR systems only make direct observations in the radial direction, usually target range, and radial velocity from Doppler frequency. Doppler resolution can be unreliable since, in most cases, target motion is uncontrolled and may not provide sufficient Doppler diversity. Rotating target features can exhibit Doppler differences, which can resolve the target or at least provide opportunities to gather information about its size, shape, and orientation. It is our intention to show that if we estimate radial acceleration from the ISAR data, we can obtain information about target geometric parameters by comparing "point scattering features" on the target and incorporating a priori knowledge/modeling constraints.

The motivation behind estimation of these parameters is to facilitate automatic target recognition (ATR) post-processing. It is our goal to provide quality inputs to an ATR system developed as byproducts of ISAR imaging. We believe through effective modeling of ISAR motion and manipulation of imagery, that target geometry and rigid body motion can be effectively estimated.

To set up the present problem, we assume that the ISAR RADAR has been cued to a target vehicle and that the grazing angle is shallow allowing a nearly horizontal line-of-sight although large enough that occlusion of target features does not limit the availability of admissible target features. Accordingly, we assume that sufficient target features are detected to create an ISAR image. We begin by deriving motions visible to an ISAR to show the relationships between identifiable target parameters and motions.

2.0 ISAR TARGET MOTION AND DISPLACEMENT

The purpose of this section is to motivate the remainder of this paper and place our estimation techniques in firm theoretical and physical footing. The primary modeling assumption that we use is that the target is a rigid body. The second assumption that we make, although not truly necessary, is that the target is composed of a countable number of strong reflecting features that we can model as point scatterers. Individual point scatterers displaced from the centroid of the target by the vector $\mathbf{\Delta R}=[r, w, z]^T$, where the components r, w, and z stand for range, horizontal, and vertical displacement in a

Further author information –
K.A.M.: Email: kam@trg.saic.com
J.R.B.: Email: jrb@trg.saic.com

Cartesian coordinate system. The grazing angle is assumed to be low such that for the purposes of this analysis the line of sight is assumed to be in the horizontal plane. The target exhibits no transversal motion since bulk motion compensation keeps the target centroid centered at the origin of our frame, but the target exhibits angular motion about the centroid according to $\mathbf{\Omega}=[\Omega_r, \Omega_w, \Omega_z]^T$. Since the range component of angular velocity is about the range axis, it causes no change in range, and consequentially, Ω_r cannot be observed via from instantaneous range and Doppler frequency measurements. To derive the radial velocity with respect to the ISAR, we place the target at rest in a rotating frame with angular velocity, $\mathbf{\Omega}$. The instantaneous velocity in the range direction can be found as[1]

$$v_r = \mathbf{e}_r \cdot (\Omega \times \Delta\mathbf{R}) = \Omega_w z - \Omega_z w, \tag{1}$$

where $\mathbf{e}_r$ is the unit vector in the range direction. The acceleration of the target can then be found as the time derivative of v_r,

$$a_r = \frac{d(\mathbf{e}_r \cdot (\Omega \times \Delta\mathbf{R}))}{dt} = \Omega_w v_z - \Omega_z v_w + \dot{\Omega}_w z - \dot{\Omega}_z w = -(\Omega_w^2 + \Omega_z^2)r + \Omega_r(\Omega_w w + \Omega_z z) + \dot{\Omega}_w z - \dot{\Omega}_z w \tag{2}$$

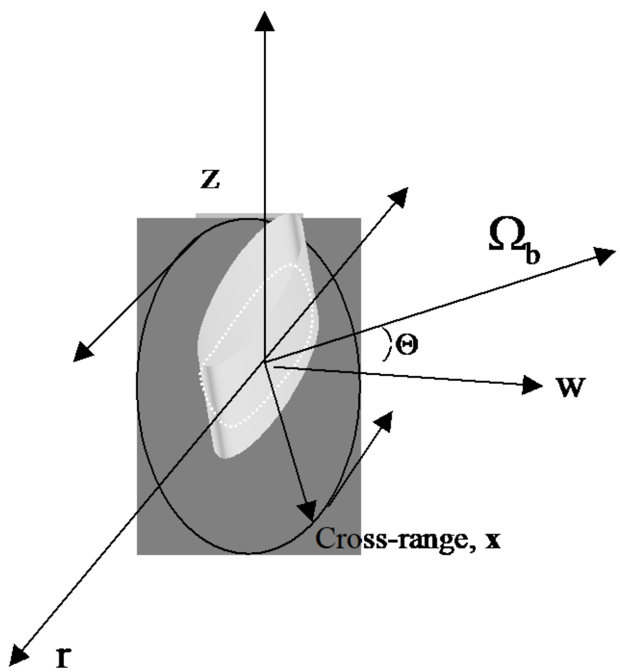

Figure 1: ISAR Projection Geometry

To continue with our analysis, we convert our expressions for acceleration to the coordinate system of the ISAR focus plane, a new rectangular coordinate system, (r, x, b). r is still the range or line-of-sight, x is the cross-range coordinate proportional to Doppler frequency, and b is the displacement perpendicular to the slant plane and unobservable in ISAR imagery. The angular motion that the ISAR can perceive is limited to the vector sum, $\mathbf{\Omega}_b = \mathbf{\Omega}_w + \mathbf{\Omega}_z$. Hence $\mathbf{\Omega}_x$ is defined to be 0. Transformation into this coordinate system is accomplished by rotating the frame by angle Θ around the range (line-of-sight) axis. Knowing the angle Θ allows the image to be projected back into the horizontal and vertical components to provide height and width, see Figure 1. In essence, the ISAR imagery is a projection of the target into the plane normal to $\mathbf{\Omega}_b$ and scaled by its magnitude.

The resulting expression for acceleration is now

$$a_r = \frac{d(\mathbf{e}_r \cdot (\Omega \times \Delta\mathbf{R}))}{dt} = -\Omega_b v_x - \dot{\Omega}_b x = -\Omega_b^2 r + \Omega_r \Omega_b b - \dot{\Omega}_b x \tag{3}$$

Using the cross-range scale factor $2\Omega_b/\lambda$ to convert cross-range to the more familiar Doppler frequency, we arrive an expression based on range and frequency,

$$a_r = -\Omega_b^2 r + \Omega_r \Omega_b b - \frac{\dot{\Omega}_b \lambda}{2\Omega_b} f \ . \tag{4}$$

The middle term is problematic since it depends on the unobservable roll Ω_r and perpendicular displacement *b*. Fortunately, it should be small for land targets. But in the case of sea or air targets, it will remain a source of error and de-focusing. The first and third terms, however, provide a method of relating physical target parameters to the derivatives of the target acceleration with respect to range and to frequency, which are respectively,

$$\frac{da_r}{dr} = -\Omega_b^2 \text{ and } \frac{da_r}{df} = -\frac{\dot{\Omega}_b \lambda}{2\Omega_b} . \tag{5}$$

Since the current SAIC ISAR processor uses a truncated Taylor expansion to approximate acceleration, the above analysis indicates that the parameters of this expansion can also be used to estimate angular rate and angular acceleration up to a sign. This has been the focus of the enhancements made to the SAIC ISAR processor and the main point of this paper. Before we can relate the details, it will be helpful to review the existing SAIC ISAR processor to understand how this expansion can be estimated.

3.0 SAIC ISAR PROCESSING

The SAIC ISAR processor[2], outlined in Figure 2, consists of a series of coordinated algorithms for identifying target scattering points, measuring advanced motion parameters, and producing movie frames. The processor is composed of two nearly independent processes, which serve to isolate and focus the target.

The first of these stages starts with the raw data and reduces it to a three-dimensional array of "Sub-Images." Sub-Images are valid, coarse resolution ISAR images, the result of a 2D FFT of a short burst of pulses and thus have a short coherent integration time. Motion compensation accounts for platform and gross target motion so that range migration is minimized during Sub-Image formation. A *track-before-detect* procedure is applied to the raw, range-compressed data stream to identify potential scatterers. Aliases are compared to resolve velocity ambiguities. Affinity grouping, based on similar velocity and constrained by final image size, is used to thin the scatterer set.

The subsequent stage converts the array of coarse resolution images to a stack of fine resolution complex images, which are highly overlapped in time. The process of combining the (movie) images into a single composite image requires proper cross-range scaling and orientation of each sub-image. Effectively, this requires knowledge of the target motion over an extended time interval. The processor develops this information in a bootstrap type of procedure through which targets identified in one frame serve to improve the focus of the following frames. This is implemented by a series of code elements which we call the "Global Motion Model" (GMM). The GMM consists of routines that receive scatterer reports from both the front-end (motion-compensation) and the back-end (frame generation) loops. All of these scatterer reports consist of an array of numbers representing the target's estimated SNR, Doppler frequency, range, acceleration (Hz/second), and Doppler width. This information is stored in a large array. Before a new complex image frame is formed, the GMM sorts through all the currently available information and builds a mathematical model of it to estimate the rate of path curvature versus range, Doppler, and time. Target acquisition stage identifies focus zone of range cells on the target. Within this zone, the Doppler centroid is estimated from intelligent weighting of the range cell with the smallest Doppler width. Estimation of the centroid is done via the signal history covariance estimate for a sliding window. Inputs to the motion model are lists of target parameters, from both target acquisition and frame generation (Phase Gradient Autofocus); target report is time, frequency, range, Doppler width, acceleration, and source. Reports are thinned to an "*affinity*" group; all reports in this thinned group are used to build an empirical model of a_r(f,r,t).

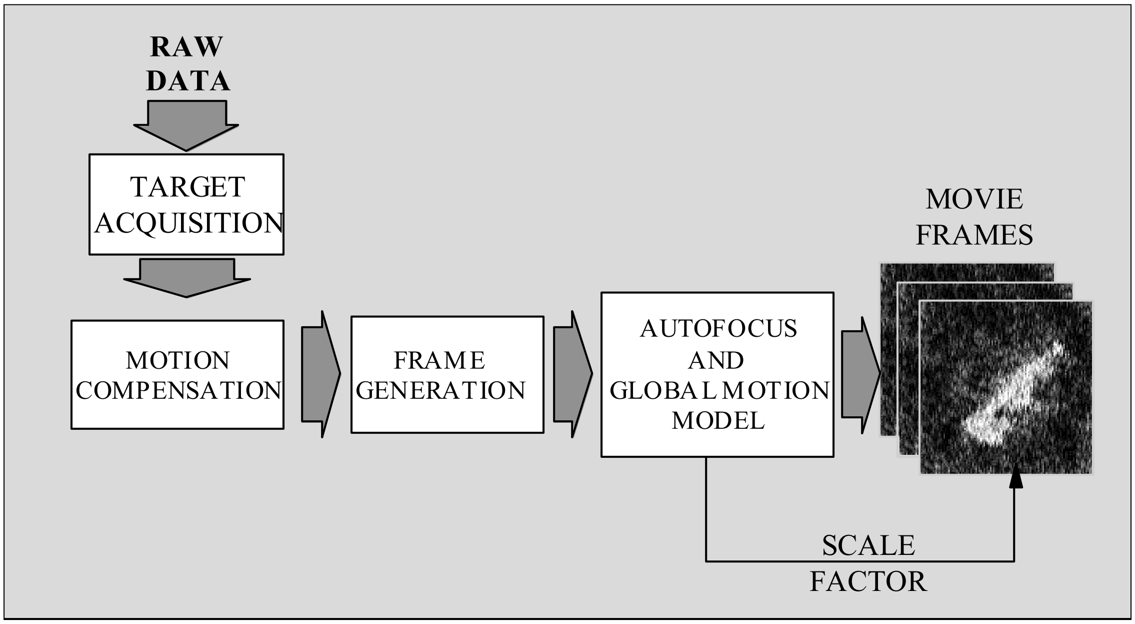

Figure 2. ISAR algorithm summary

The last process to generate the output imagery and correct via the Phase Gradient Autofocus (PGA) technique. The PGA algorithm is used to estimate and remove all phase errors that are independent of both the range and cross-range coordinates.

3.1. GLOBAL MOTION MODEL

The goal of the global motion model is to estimate the acceleration field parameterized by Doppler frequency, range, and time. Our approach is to derive a least-squares fit based upon the values of target point scatterer estimated accelerations and positions. It can be derived by estimating the acceleration field of the global target via a Generalized Taylor Expansion truncated to finite terms around the centroid location. We assume that the acceleration field can be approximated by a polynomial of range, Doppler frequency, and time,

$$\begin{aligned}\hat{a}(r,f,t) &= a(r_0,f_0,t_0) + \left.\frac{\partial a}{\partial r}\right|_{(r0,f0,t0)}(r-r_0) + \left.\frac{\partial a}{\partial f}\right|_{(r0,f0,t0)}(f-f_0) + \left.\frac{\partial a}{\partial t}\right|_{(r0,f0,t0)}(t-t_0) \\ &\quad + \left.\frac{\partial^2 a}{\partial r \partial t}\right|_{(r0,f0,t0)}(r-r_0)(t-t_0) + \left.\frac{\partial^2 a}{\partial f \partial t}\right|_{(r0,f0,t0)}(f-f_0)(t-t_0) \\ &= a_0 + a_r r + a_f f + a_t t + a_{rt} rt + a_{ft} ft \\ &\mathbf{u}\boldsymbol{\alpha}\end{aligned} \tag{6}$$

where $\mathbf{u} = [1, r, f, t, rt, ft]$, $\boldsymbol{\alpha} = [a_0, a_r, a_f\, a_t, a_{rt}, a_{ft}]^T$, (t_0, r_0, f_0) are the time, range, and Doppler at the target centroid. We can minimize this approximation with respect the mean squared error (MSE) cost criteria. The standard least-squares solution for N current target reports (the number assumed to be present during the integration time) comes simply by solving the linear equations to minimize

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}} \sum_{n=1}^{N} \frac{(y_n - \mathbf{u}_n \boldsymbol{\alpha})^2}{dw_n^2} \tag{7}$$

where y_n is the estimate of $a_r(r,f,t)$ acceleration of the nth target scattering point and dw_n is its Doppler width. The set of linear equations is solved via QR expansion and the acceleration field is calculated based upon the coefficients, α.

3.2 SEQUENTIAL ESTIMATION OF ACCELERATION COEFFICIENTS

The algorithm, described in the previous section, based estimates of the acceleration coefficients and hence the acceleration field on a least squares fit of the acceleration of current set of point target scatterers to the above global motion model. It is obvious in that a number of events may ruin this fit and acceleration calculation. Clutter or sidelobe false targets may contaminate our list of target features, the target features present may result in a singular solution, or there simply may be too few detected target features. The focus of the final image frame may be compromised if based only on the set of target scattering points detected during the integration time.

The above algorithm was extended to use memory of past coefficient fits to allow us to coast the coefficients through trouble spots and smooth the fluctuation of coefficients from image frame to frame. Indeed, the Kalman filter that we have incorporated not only provides smoothed tracking of coefficient estimates but also reinterprets the elements of the model in familiar sequential parameter estimation terms. We can incorporate the temporal relationships between coefficients $\{a_0, a_r, a_f\}$ and $\{a_t, a_{rt}, a_{ft}\}$, respectively into our model and are able to tradeoff the values of past coefficient estimates with the information derived from the current target list separately.

The variant of Kalman filter implemented is popularly known as the *information filter*[3]. So named because instead of recursively estimating a state error covariance matrix, the information filter propagates the inverse or information matrix (information in the *Cramer-Rao* sense). This sequential estimator was chosen for primarily two reasons: 1) Since all targets in the current target list are used each update/frame, the information filter is used to reduce the dimension of matrices to be inverted from N, the number of targets, to 3, the number of independent eigenmodes. 2) Initialization of the information filter begins naturally with a zero information matrix. The coefficient vector α was chosen as the state vector and the other model matrices along with definitions are specified in Table 1.

The Kalman information filter consists of a cascaded set of equations used to recursively update states, gains, and information matrices at each block update. Here, we summarize the kth iteration of the algorithm in five steps shown in Figure 3:

1. The noiseless state prediction information matrix, $A(k)$, is an intermediate result found by applying the state transition matrix to the error covariance, $P(k|k)$.
2. The state prediction information matrix, $P(k|k-1)^{-1}$, can be found by incorporating the process noise. This version of the equation follows from the matrix inversion lemma.
3. Finally, the state information matrix is found by applying the new information found as $H(k)R(k)^{-1}H(k)$.
4. The Kalman gain $W(k)$ follows by definition.
5. The state update equation is based on the standard Kalman filter implementation.

The Kalman information filter propagates two quantities: the state information matrix and the state vector. Initialized as a zero matrix, the information matrix indicates that current state estimates should have no value relative to be update. The Kalman gain is then found to be large limited by the weighting of expected measurement error in the R matrix. It grows as iteration progresses reducing the Kalman gain and inclusion of new information to a steady state level. This level is dictated by the relative weighting placed of current new updates via R and by Q, the process noise matrix which limits the size of the information matrix and our confidence in past estimates. A large value of an eigenvalue of Q indicates that the state values associated with that mode are non-stationary and vary rapidly from iteration to iteration. For instance, if we assume that the mean value of acceleration changes too much from frame to frame to be predictable, we set σ_I^2 to an arbitrarily large value. The Kalman gains associated with this mode will be zero and the state variables associated with this mode will not be tracked. This is useful for selectively eliminating or downplaying portions of the model.

NAME	DEFINITION	SIZE	VALUES

x	State vector – coefficients to be estimated	1x6	$\begin{bmatrix} a_0 & a_r & a_f & a_t & a_{rt} & a_{ft} \end{bmatrix}^T$
Φ	State Transition matrix – describes the coupling between state variables and their propagation in unforced conditions.	6x6	$\begin{bmatrix} 1 & 0 & 0 & T & 0 & 0 \\ 0 & 1 & 0 & 0 & T & 0 \\ 0 & 0 & 1 & 0 & 0 & T \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
H	Observability matrix – weights the linear combination of states to estimate acceleration	Nx6	$\begin{bmatrix} 1 & \cdots & 1 & \cdots & 1 \\ r_1 & \cdots & r_n & \cdots & r_N \\ f_1 & \cdots & f_n & \cdots & f_N \\ t_1 & \cdots & t_n & \cdots & t_N \\ r_1 t_1 & \cdots & r_n t_n & \cdots & r_N t_N \\ f_1 t_1 & \cdots & f_n t_n & \cdots & f_N t_N \end{bmatrix}^T$
R	Measurement Noise Matrix – estimates our ability to accurately measure acceleration, and relative value of updates. Here, we have naturally used the Doppler width as the standard deviation of the acceleration measurement error. Note that target reports are assumed to be independent.	NxN	$\begin{bmatrix} dw_1^2 & 0 & \cdots & 0 & 0 \\ 0 & \ddots & 0 & 0 & 0 \\ \vdots & 0 & dw_n^2 & 0 & \vdots \\ 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & \cdots & 0 & dw_N^2 \end{bmatrix}$
Q	Process Noise/Maneuver Matrix - acceleration sequence treated as white noise associated with the following modes: the spatial average, range, and Doppler frequency.	3x3	$\begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_r^2 & 0 \\ 0 & 0 & \sigma_f^2 \end{bmatrix}$
Γ	Process Noise Gain – assumes a piecewise constant white noise acceleration model and temporal coupling of noise modes.	6x3	$\begin{bmatrix} T^2/2 & 0 & 0 \\ 0 & T^2/2 & 0 \\ 0 & 0 & T^2/2 \\ T & 0 & 0 \\ 0 & T & 0 \\ 0 & 0 & T \end{bmatrix}$

Table 1. Information Filter Matrix Quantities

Calculate the Noiseless State Prediction Information Matrix

$$A(k-1) = [\Phi^{-1}]^T P(k-1|k-1)^{-1}\Phi^{-1}$$

$\downarrow$

Calculate the State Prediction Information Matrix

$$P(k|k-1)^{-1} = A(k-1) - A(k-1)\Gamma[\Gamma^T A(k-1)\Gamma + Q^{-1}]^{-1}\Gamma^T A(k-1)$$

$\downarrow$

Update the State Information Matrix

$$P(k|k)^{-1} = P(k|k-1)^{-1} + H(k)^T R(k)^{-1} H(k)$$

$\downarrow$

Calculate the Kalman Gain

$$W(k) = P(k|k)H(k)^T R(k)^{-1}$$

$\downarrow$

Update the State Vector

$$x(k) = \Phi x(k-1) + W(k)[y(k) - H(k)x(k-1)]$$

Figure 3. Information Filter Algorithm

4.0 ESTIMATION OF TARGET PHYSICAL PARAMETERS

Using the estimation techniques derived in Section 3 to solve for the acceleration derivatives derived in Section 2, we find that we can estimate the global angular velocity and acceleration of a rigid target body. This information is sign ambiguous in the sense that the function $\Omega_b(t)$ could be replaced by $-\Omega_b(t)$. Once a sign is chosen for one time, the variation with time is unique. Since the cross range coordinate and stretching are defined by these quantities respectively, we find that we can appropriately scale the ISAR imagery to estimate the physical dimensions of the target projected into the target focus plane. In instances where target motion can be specified as pitch or yaw only, target heights or widths can be determined. When both motions are acting, a slantwise slice is shown in the imagery. Conversion of scatterer points into physical coordinates then requires rotation by the angle resulting from the vector sum pitch and yaw components, Θ.

4.1 ESTIMATION OF ANGULAR VELOCITY, ANGULAR ACCELERATION, AND CROSS-RANGE DISPLACEMENT

Estimation of the absolute value of the rotation rate follows simply from

$$\left|\hat{\Omega}_b(t)\right| = \sqrt{\left|a_r + a_{rt}(t-t_0)\right|}\,, \tag{8}$$

assuming that the vehicle is a rigid body and the motion around the centroid is purely rotation. This rotation rate is proportional to the cross-range scale factor, the cross-range distance from the centroid can be found as $(f\text{-}f_0)\lambda/2|\Omega_b|$. Again, this is the minimum distance from the scatterer to the $\mathbf{\Omega}_b$ axis of rotation. In limited cases, we may be able to apply the cross-range distance to find actual target dimensions. For instance in the case of a ship at a 0° or 180° heading to the ISAR experiencing pure pitch due to wave motion ($\mathbf{\Omega_b} = \mathbf{\Omega}_w$), the distance measured by the cross range coordinate is normal to Ω_w or in the *z* direction. The resulting ISAR imagery would be show the ship projected into the vertical plane. Similarly, a turning ship experiencing only a yaw component might provide an ISAR image of the ship projected into the horizontal plane. In most cases, the resultant focus plane is a slantwise cut through the target, which nevertheless provides some size and shape information.

Additionally, the angular acceleration can be estimated directly from a_f via

$$\frac{d\hat{\Omega}_b}{dt} = \frac{2(a_f + a_{ft}(t - t_0))\left|\hat{\Omega}_b\right|}{\lambda} \tag{9}$$

Determination of the sign of Ω_b is more difficult since it cannot directly be determined from the coefficients and is currently a topic of research. In itself $d\Omega_b/dt$ may be used to de-stretch target cross-range coordinates.

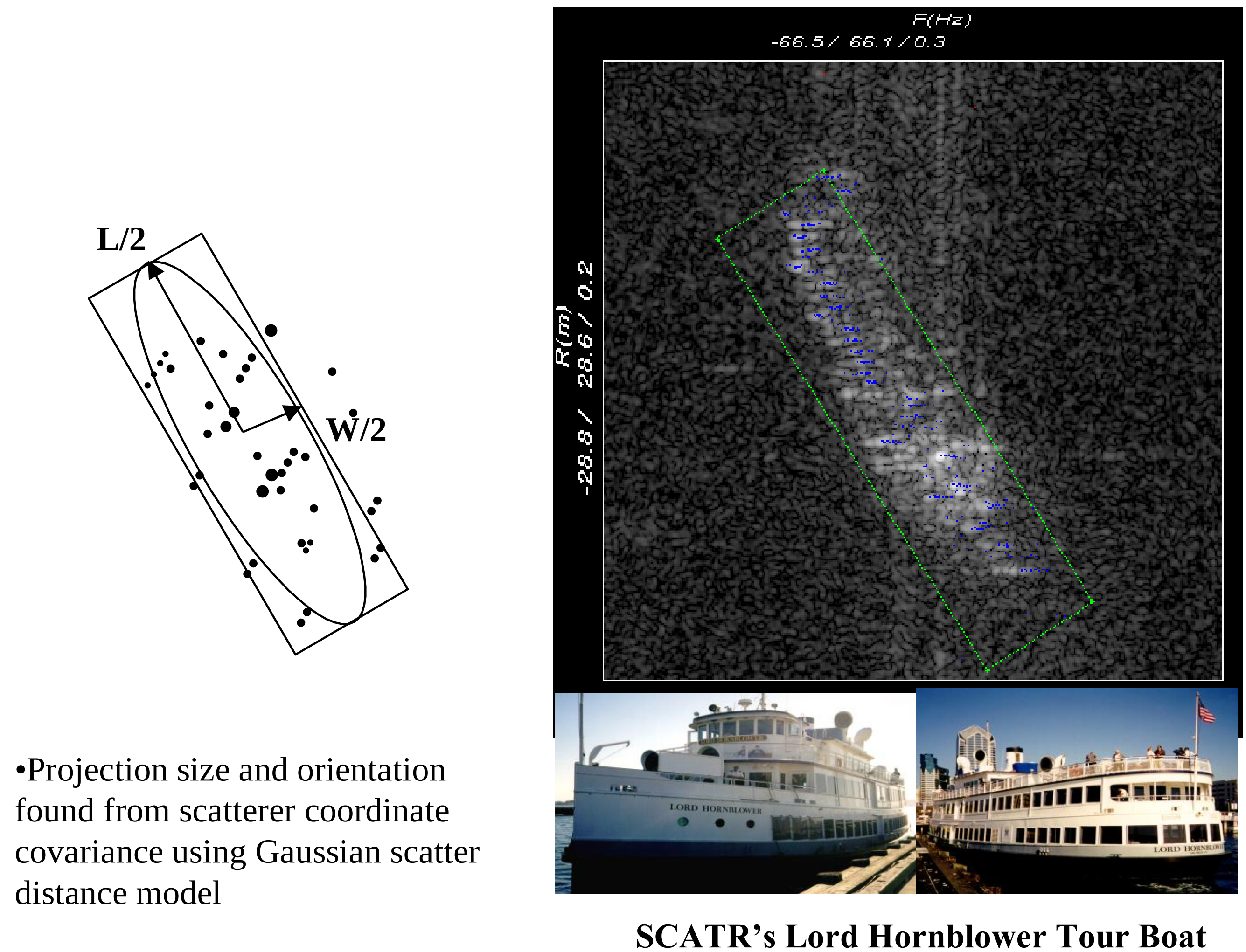

Figure 4. Illustration of target projection size calculation

4.2 ESTIMATION OF TARGET COORDINATES AND ORIENTATION

An ambitious and perhaps impractical goal for parameter estimation to support ATR processing would be to provide 3-D coordinates of individual point scatterers that compose the ISAR target. The chief difficulty is that the ISAR represent the target as a 2-D projection of the target at an angle determined by the relative magnitudes of pitch and yaw. Estimation of this angle Θ is the beyond the scope of this note, but we can briefly describe what is necessary for this task. Since rotation around the line-of-sight is ambiguous to the RADAR, determination of any absolute angle is not possible without *a priori* knowledge. However, measurement of the relative angle between slices may be accomplished by comparison of frames. Correct estimation of relative angles between frames and proper scale may lead to 3-D representations of target scattering points.

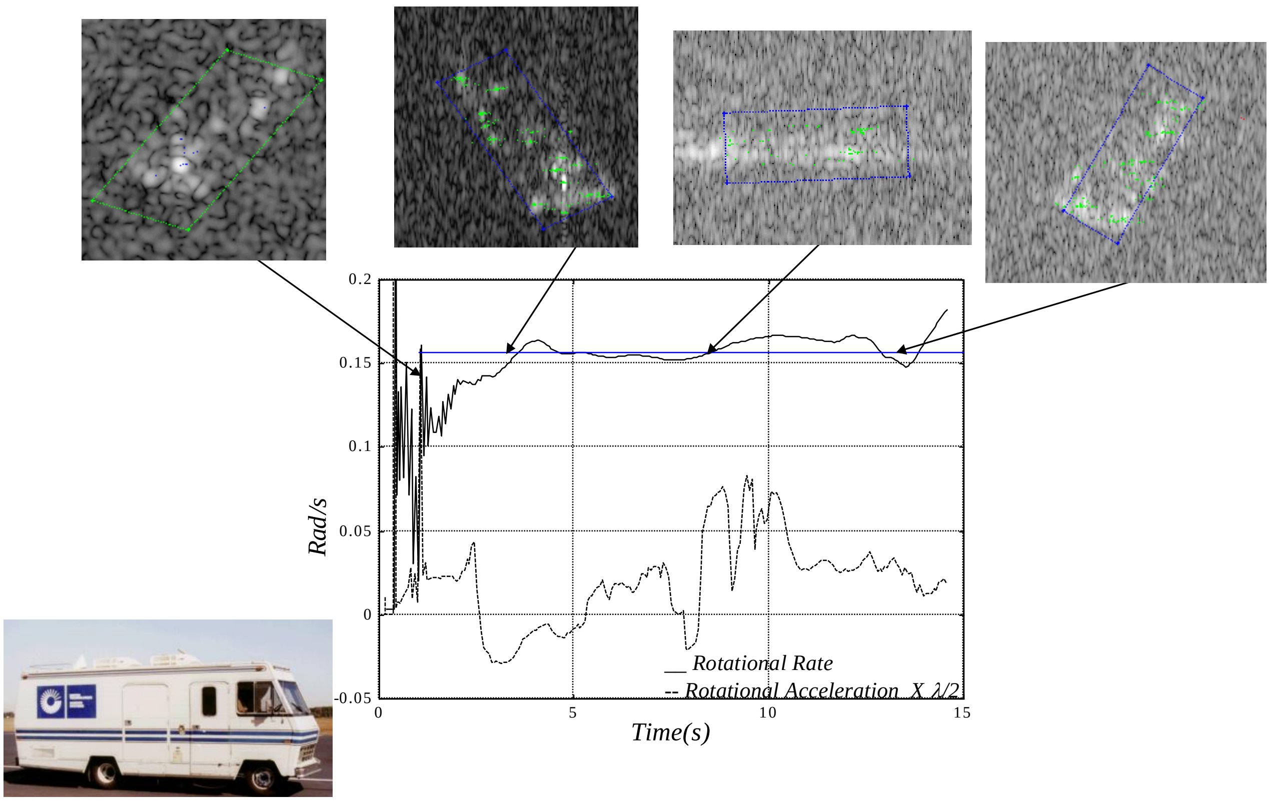

Figure 5. Norden Winnebago Motor Home Rotational Parameters
0.88 sec. Integration Time, 188 256x256 frames

In this note we do not try to determine the angle of the target projection but instead look to estimate the target scatterer coordinates within the slant plane. From the previous estimates of angular velocity and acceleration we are able to scale and de-skew the cross range coordinates of individual target scatterers. The final basic processing that can be done is determining the approximate orientation and size of the target within the focus slant plane. The parameters of an ellipse can be calculated if we assume a statistical Gaussian scatter model. This follows simply from calculation of the sample covariance matrix of the coordinates of target points in the slant plane. The square roots of the eigenvalues of this matrix are the major and minor axes of an ellipse while the eigenvector matrix (a matrix whose columns are eigenvectors) is a rotational matrix providing the target orientation. The basic model used in the SAIC ISAR assumes the target fits a rectangle with dimensions of twice the major and minor axis lengths under the same rotation, see Figure 4. Inscribing the target in a box is motivated by the secondary purpose of the identifying target scatterer reports which appear outside of the rectangular confines of the target and removing them from consideration in deriving model coefficients. We find this approach performs well in practice except in when sidelobes of bright target scatterer points are detected by PGA and added to the target scatterer report list. Naturally, the target box enlarges to incorporate these additional target reports. Provisions have also been made to override the estimates made from individual scatterers to employ *a priori* information of length, width, and orientation. In many cases such as land targets the motion the target must undergo is predictable and may used instead of noisy estimates from data.

5.0 RESULTS AND ANALYSIS

To illustrate and reinforce the above concepts we offer three pertinent examples of ISAR processing supplemented with target parameter estimation. The initial case is that of a controlled experiment shown primarily to validate some of the concepts with experimental evidence. It consists of a Winnebago motor home turning at a fixed rate. The remaining two examples provide typical examples of ISAR imaging of small naval craft. Small craft were chosen because they represent a serious challenge to an ISAR processor. For instance, they may be composed of fewer and dimmer scatterers and may well exhibit more rapid fluctuations in motion. The first point leads us to believe that it may be more difficult to make accurate estimates of motion for small craft, while the second indicates that it may be harder to track the motion of small craft.

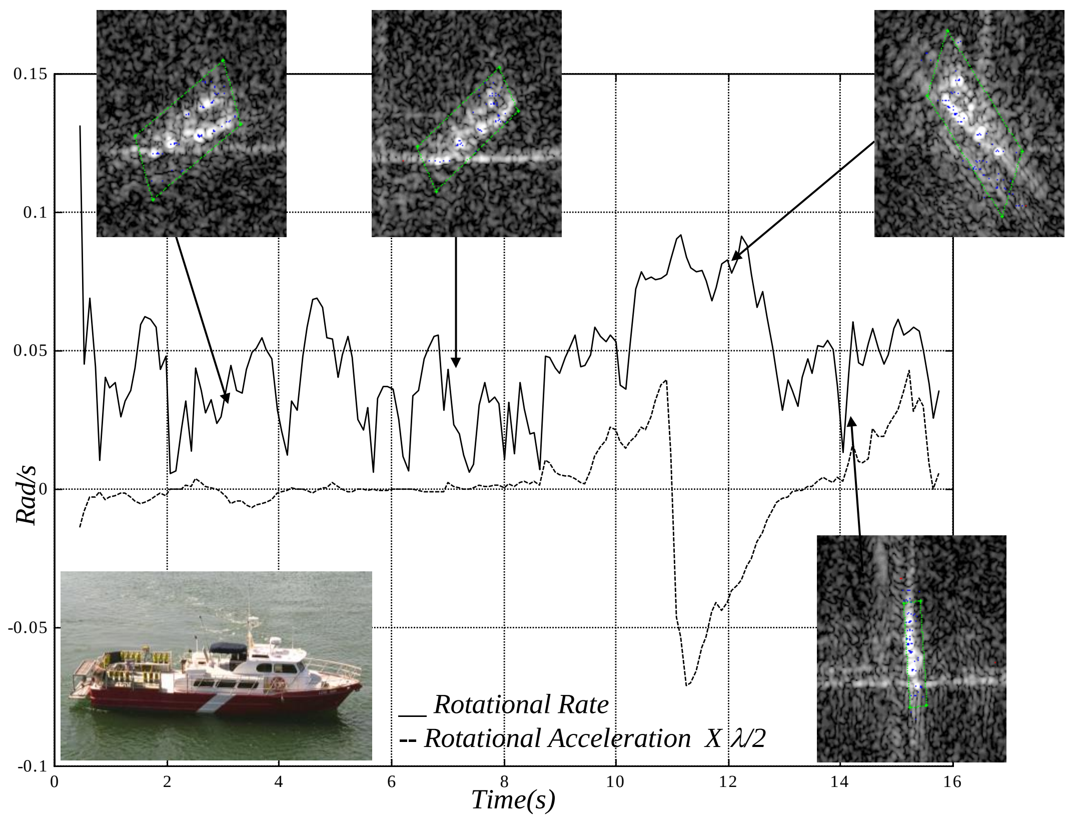

Figure 6. "Loise Ann" Dive Boat Rotational Parameters
47' length, 0.72 sec. Integration Time, 172 256x256 frames

The "Winnebago" data was obtained from Northrop Grumman Norden systems collected on a AN/APG-76 Ku-Band, three aperture radar. The radar collects three channels of data, which can be used to produce a clutter-cancelled channel that was used. The Winnebago motor home is shown in Figure 5 was slightly altered by the addition of a triangular corner reflector mounted on the roof above the logo. The vehicle was imaged in a turn executed with a fixed rate of approximately 150 mrad/sec over flat ground. Figure 5 shows the result of imaging 11.5 seconds of data to generate 188 image frames each with an integration time of about .88 seconds. Not that our estimates of Ω_b and $d\Omega_b/dt$ track the 150 mrad/s nominal value. It should also be noted that from about 6.5 seconds to 9.5 second, clutter cancellation canceled the target. Using the Kalman filter, our estimates were able to coast over this loss of target scatterers. Note that the target was "boxed" with calculated length, width, and orientation values for the image nominally at 1.5 sec. The remaining three images defaulted to a priori values of length, height, width, and orientation to draw the target box. Target scatterers used in acceleration and box drawing calculations are also marked in the image.

The remaining examples use data from the ONR's Small Craft ATR (SCATR) program's San Diego collection. In particular, we process data of a 47' dive boat "Loise Ann" and the Coast Guard Cutter Tybee, which were collected using a X-Band RADAR with 200 Hz PRF and 500 MHz bandwidth (1 foot resolution). Figure 6 shows 15.6 seconds of dive boat data that was processed rendering 172 frames with a .72 second integration time. The target makes long slow turns apparently free of wave motion. Estimates of $d\Omega_b/dt$ appear consistent with estimates of Ω_b, and estimates of length, width, and orientation shown by the box inscribing the target appear well behaved.

The final example, shown in Figure 7, consists 25 seconds of the data of the Coast Guard cutter "Tybee," processed to render 312 frames with a .64 sec. integration time. The reason why the integration time is so short is that the Tybee appears to be undergoing periodic wave motion, which modulates the motion of its turns. This evident from the rapid swinging of its central mast from positive to negative Doppler frequency. Likewise the cross-range resolution fluctuates as the target undergoes angular acceleration. When the cross-range scale factor is very small, the lack sufficient spread of Doppler frequencies may cause the least-square problem to become ill conditioned leading to poorer estimates of the acceleration coefficients and, therefore, Ω_b and $d\Omega_b/dt$. In addition, the Tybee experiences yaw indicated by the tilt of the ship image in the image taken at 6.8 sec. In these instances rotation rate remains nonzero for several seconds. In this case in particular, we find that yaw generates a slow changing Ω_b similar to the previous example, but one which is modulated by rapid pitching motion of pitch.

6.0 SUMMARY AND CONCLUSIONS

In this note, we derived rotational parameters and scale factors for a rigid ISAR target based on the by-products of ISAR processing, or more specifically acceleration estimation techniques used for focusing. We provided initial results for simple land targets and small naval craft. Other accomplishments include improved parameter tracking via incorporation of a Kalman filter, improvements to our ability to extract target features by elimination of clutter and sidelobes inconsistent with the assumed target slice shape. Lastly, a physical interpretation for the acceleration field estimated in the global motion model has been devised. We are encouraged by the promise of this approach and look to extend our parameter estimation techniques to include: 1) the relative sign of Ω_b, 2) the relative angle between image slices, 3) parameter maneuver models for land and sea, and 4) further validation of our parameter estimates based on the ground truth of collected data.

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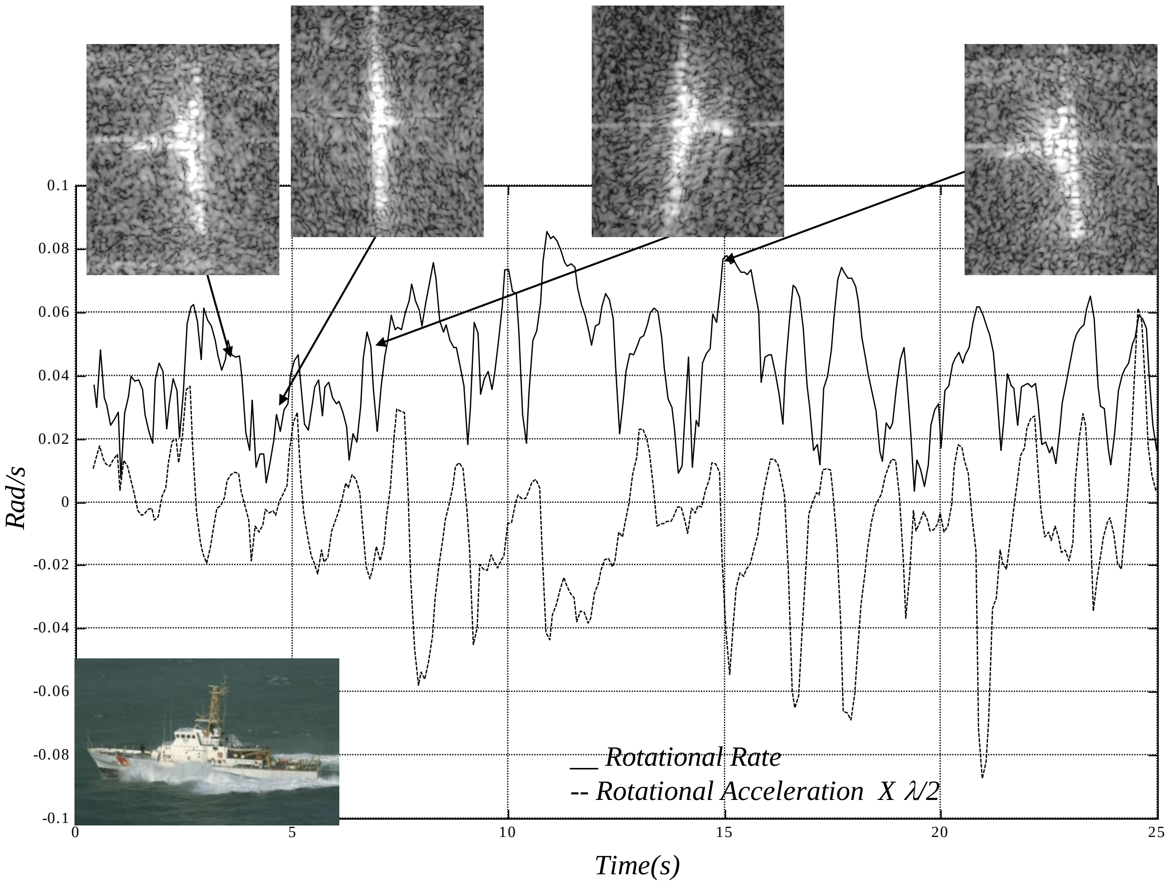

Figure 7. US Coast Guard Cutter Tybee (WPB-1330) Rotational Parameters
110’ Patrol Boat, 21’ beam, 0.64 sec. Integration Time, 312 256x256 frames